

Practice Midterm

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Problem 1 You throw a fair die six times in a row. What is the probability that all of the six face values appear? What is the probability that one or more sixes appear?

Solution:

$$\mathbb{P}(\text{All six faces appear}) = \frac{6!}{6^6} = 0.0154 \quad (1)$$

$$\mathbb{P}(\text{one or more sixes appear}) = 1 - \mathbb{P}(\text{no sixes appear}) = 1 - \left(\frac{5}{6}\right)^6 = 0.6651. \quad (2)$$

Problem 2 Your friend and her husband did a 100% reliable sonogram test and found out that they are going to have twin boys. They want to know whether their boys are identical twins or fraternal twins. (For your information, identical twins are always of the same sex, but the sexes of fraternal twins can be different. Moreover, you can assume that among the identical twins, it is equally likely that the twins are both boys or both girls; and that among the fraternal twins, the sexes of a random pair of twins are independent and equally likely to be male or female.) The doctor only told them that out of the twin population, the proportion of identical twins is one third. What is the probability that the couple's boys will be identical twins? Solution:

$$\begin{aligned} & \mathbb{P}(\text{identical twins} | \text{twin boys}) \\ &= \frac{\mathbb{P}(\text{identical twin boys})}{\mathbb{P}(\text{twin boys})} \\ &= \frac{\mathbb{P}(\text{identical twins})\mathbb{P}(\text{identical twin boys} | \text{identical twins})}{\mathbb{P}(\text{identical twins})\mathbb{P}(\text{identical twin boys} | \text{identical twins}) + \mathbb{P}(\text{fraternal twins})\mathbb{P}(\text{fraternal twin boys} | \text{fraternal twins})} \\ &= \frac{1/3 \times 1/2}{1/3 \times 1/2 + 2/3 \times 1/4} \\ &= 1/2 \end{aligned} \quad (3)$$

Problem 3 Suppose that you wish to invest in two stocks A and B. The rates of return on these stocks in the coming year depend on the development of the economy. The economic prospects for the coming year consists of three equally likely case scenarios: a strong economy, a normal economy, and a weak economy. If the economy is strong, the rate of return on stock A will be 34% and that on stock B will be -20%. If the economy is normal, the numbers are 9.5% for stock A and 4.5% for stock B. If the economy is weak, they are -15% for stock A and 29% for stock B.

- Compute the expected returns on stocks A and B.
- Compute the correlation between the returns on stocks A and B.
- Suppose you invest a half (of your investment money) in stock A and half in stock B. Calculate the expected value and standard deviation of your overall return.

Solution: a) Let X_A be the return of stock A, X_B be the return of stock B, in the coming year.

$$\mathbb{E}[X_A] = \frac{1}{3}(0.34 + 0.095 - 0.15) = 0.095 \quad (4)$$

$$\mathbb{E}[X_B] = \frac{1}{3}(-0.20 + 0.045 + 0.29) = 0.045 \quad (5)$$

b)

$$\mathbb{E}[X_A X_B] = \frac{1}{3}(-0.34 \times 0.20 + 0.095 \times 0.045 - 0.15 \times 0.29) = -0.0357. \quad (6)$$

$$Cov[X_A X_B] = \mathbb{E}[X_A X_B] - \mathbb{E}[X_A]\mathbb{E}[X_B] = -0.04. \quad (7)$$

$$\mathbb{E}[X_A^2] = \frac{1}{3}(0.34^2 + 0.095^2 + 0.15^2) = 0.049 \quad (8)$$

$$\mathbb{E}[X_B^2] = \frac{1}{3}(0.20^2 + 0.045^2 + 0.29^2) = 0.042 \quad (9)$$

$$Var(X_A) = \mathbb{E}[X_A^2] - (\mathbb{E}[X_A])^2 = 0.04 \quad (10)$$

$$Var(X_B) = \mathbb{E}[X_B^2] - (\mathbb{E}[X_B])^2 = 0.04 \quad (11)$$

$$Corr(X_A, X_B) = \frac{Cov(X_A, X_B)}{\sqrt{Var(X_A)}\sqrt{Var(X_B)}} = \frac{-0.04}{\sqrt{0.04}\sqrt{0.04}} = -1. \quad (12)$$

c)

$$\mathbb{E}[0.5X_A + 0.5Y_B] = 0.5\mathbb{E}[X_A] + 0.5\mathbb{E}[X_B] = 0.07 \quad (13)$$

$$Var(0.5X_A + 0.5X_B) = 0.5^2 Var(X_A) + 0.5^2 Var(X_B) + 2 \times 0.5^2 Cov(X_A, X_B) = 0 \quad (14)$$

Problem 4 An airline knows that 5 percent of the people making reservations on a certain flight will not show up. Consequently their policy is to sell 52 tickets for a flight that can hold only 50 passengers. What is the probability that there will be a seat for every passenger who shows up?

Solution: We are interested in $\mathbb{P}(X \leq 50)$, i.e., the probability that there is a seat for all passengers that show up.

$$\begin{aligned} \mathbb{P}(X \leq 50) &= 1 - \mathbb{P}(X > 50) \\ &= 1 - \mathbb{P}(X = 51) - \mathbb{P}(X = 52) \\ &= 1 - 52(0.05)(0.95)^{51} - (0.95)^{52} = 0.74 \end{aligned} \quad (15)$$

In the above, the second term represents the probability that exactly 51 people show up which includes the possibilities that any 1 of the 52 persons does not show up and the rest show up, and the last term represents the probability that all 52 persons show up.

Problem 5 Let $X_1, X_2, \dots$ be a sequence of independent identically distributed continuous random variables. We say that a record occurs at time n if $X_n > \max(X_1, \dots, X_{n-1})$. That is X_n is a record if it is larger than all the previous values. Let Y_n be the number of record occurred by time n (for clarity, say there is always a record at time 1). Find the mean of Y_n for any n , and variance of Y_n when $n = 2, 3$.

Solution: Let I_i be the indicator function of whether there is a record breaker at position i . Specifically, $I_i = 1$ if a record occurs at position i , and $I_i = 0$ otherwise. Notice that I_i equals to 1 by definition. Thus $Y_n = \sum_{i=1}^n I_i$ and

$$\begin{aligned} \mathbb{E}[Y_n] &= \sum_{i=1}^n \mathbb{E}[I_i] = \sum_{i=1}^n \mathbb{P}(I_i = 1) \\ &= \sum_{i=1}^n \mathbb{P}(\text{the } i\text{th position contains the largest value of first } i \text{ outcome values}) \\ &= \sum_{i=1}^n 1/i. \end{aligned} \quad (16)$$

On the other hand,

$$\begin{aligned} Var(1 + I_2) &= Var(I_2) = \mathbb{E}[I_2^2] - (\mathbb{E}[I_2])^2 \\ &= \mathbb{E}[I_2] - (\mathbb{E}[I_2])^2 \\ &= 1/2 - (1/2)^2 = 1/4, \end{aligned} \quad (17)$$

$$\begin{aligned}
\text{Var}(1 + I_2 + I_3) &= \text{Var}(I_2 + I_3) \\
&= \text{Var}(I_2) + \text{Var}(I_3) + 2\text{Cov}(I_2, I_3) \\
&= 1/2 - (1/2)^2 + 1/3 - (1/3)^2 + 2(\mathbb{E}[I_2 I_3] - 1/2 * 1/3)
\end{aligned} \tag{18}$$

However, Note that $I_2 I_3$ is a Bernoulli variable, with value 1 if both $I_2 = 1$ and $I_3 = 1$. The success probability of this variable is $P(X_1 < X_2 < X_3) = 1/3!$. Hence $E[I_2 I_3] = 1/3! = 1/6$ and $\text{Var}(1 + I_2 + I_3) = 1/4 + 2/9 = 17/36$.

Problem 6 A random variable X , which represents the weight (in ounces) of an article, has density function given by $f(x)$:

$$\begin{aligned}
f(x) &= (x - 8) \quad \text{for } 8 \leq x \leq 9 \\
f(x) &= (10 - x) \quad \text{for } 9 \leq x \leq 10 \\
f(x) &= 0 \quad \text{otherwise}
\end{aligned} \tag{19}$$

The manufacturer sells the article for a fixed price of \$2.00. He guarantees to refund the purchase money to any customer who finds the weight of his article to be less than 8.25 oz. His cost of production is related to the weight of the article by the relation $x/15 + 0.35$. Find the expected profit per article.

Solution:

$$\begin{aligned}
\mathbb{E}[\text{Profit}] &= - \int_8^{8.25} (x/15 + 0.35)f(x)dx + \int_{8.25}^{10} (2 - x/15 - 0.35)f(x)dx \\
&= - \int_8^{8.25} (x/15 + 0.35)(x - 8)dx + \int_{8.25}^9 (2 - x/15 - 0.35)(x - 8)dx + \int_9^{10} (2 - x/15 - 0.35)(10 - x)dx \\
&= -0.028 + 0.5016 + 0.514 = 0.9876.
\end{aligned} \tag{20}$$